

AVDASI2 2024-25 Team 08 Technical Report

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1. Executive Summary

Albatross Aviation, was tasked with creating a UAV capable of rapidly transporting blood samples for emergency purposes. The UAV had to be able to take off from short rural tarmac runways and fly to designated test centres, while its return journey by road was considered noncritical. After inspecting the requirements and specifications, it was realized that the project was subjected to wind tunnel experiments to gather aerodynamic data and an ultimate loading test; these results were then compared against theoretical calculations. The project was split into five major milestones to ensure quality control, and adherence to the timeline. These stages included the initial concept phase, a prototype review (which features the graphite goose test), the design planning phase, the final build inspection, and finally the data repository freeze. Feedback at each milestone was used by Team 8 to continuously refine and improve subsequent stages of the project.

For the design phase of the project, the team used down selection methods to decide on the most optimal fairing designs. During the down selection the team agreed on 3 types of fairings and to remain objective these selections were made by using weighted comparison matrices. The design specialists were also tasked with creating the joints, which were highly constrained by the dimensions and designs of other subassemblies, resulting in a highly iterative process. This also led to a need to constantly change designs and thus the team opted to use 3D printing as the main manufacturing method. Notably, due to the complex shape of the fairing, it had to be printed and assembled segment by segment via post-processing methods, and the empennage joint required brass inserts for assembly. The team encountered setbacks like 3D printing errors, constant adjustments to incorporate for the assembly sequences, and the need to communicate with other subassemblies to make frequent changes to the details. However, through optimization and collaboration with other sub-systems was achieved, especially the empennage section interface performed well in terms of fit and functionality.

Moving onto the structures component, the joints performed well in the theoretical design calculations and in the ultimate load tests. Some of the joints were reinforced with aluminum plates in order to increase the load carrying capacity of the PLA material, since PLA is not a good load bearing material. The center bar and fixture showed good structural stability in extreme load tests up to 813 N. The only significant failure occurred in the main wing clamp due to shear damage of the 3D printed part. The calculated Reserve Factor shows that the design is overall sound and validates the prediction of the weakest link at the beginning of the design (although no real failure occurred).

The aerodynamic function of the fairing is mainly to reduce form drag and surface friction drag by optimizing the shape at the structural interface. Although CFD analysis could not be performed due to tool and technology limitations, the team designed a fairing that conforms to the shape of the NACA2414 airfoil to maintain laminar flow and reduce turbulence, conforming to relevant aerodynamic theory. In wind tunnel tests, the increase in total drag caused by the fairing installation was very small. Hence, the streamlined shape of the fairing was able to improve flow such that the increase in surface area would not lead to significantly larger drag. Despite the limited test conditions (e.g. testing fuselage and wings separately), the results show that the fairing can be effective in improving aerodynamic performance in real flight.

The project managers of Team 8 had a heavy role involving constant inter-divisional communication. Organization documents proved useful to manage and allocate working hours, especially when faced with a certain amount of lab hours available. Improvements have been suggested to make the process smoother in future years.

To conclude, Team 8 collaborated with all other subassemblies to produce a UAV that performed well in the wind tunnel and structural testing. Looking back, the team has a deeper understanding of the assigned role and have grown from the lessons learned throughout the course of the project.

2. Group Information

This report demonstrates the design, fabrication, and evaluation process of the joints and fairings by Company A's fuselage division. We were primarily responsible for the transition of the fairing and the connection between the main structures of the joints and therefore needed to work with multiple teams to assemble it.

The primary responsibilities of the Joints and Fairings group include the design development of the UAV fuselage fairings and joints. This work included initial CAD modeling, 3d printing and fabrication, as well as conducting multiple experiments to test the feasibility of each joint at the structural level and ultimately analyzing the results.

At the company level, our role is to fabricate every major connector, perform cross-team project management and CAD design integration. Additionally, the teams collaborated on Graphite Goose testing, structural experiments, and wind tunnel testing to ensure full access to critical aircraft data and company-wide data consistency analysis.

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3. Design and Manufacturing Review

3.1 Down selection:

From the requirement specification document, specifically section 2.6.1.a, we knew that our fairing needed to cover the gaps and sharp edges created by the main wing joint and the empennage joint to avoid flow separation that would increase drag. Additionally, as outlined in 2.6.2.a, we recognized that the fairing could be made out of either Styrofoam, plastic mouldings or 3D printed plastics. Moreover, the fairing had to cope with aerodynamic loads, be lightweight and needed to be easy to assemble / disassemble. To achieve the best results possible, our team decided to use a structural approach to evaluate possible design options. Firstly, our design and aerodynamic teams met and wrote down the most important features that our fairings should have. We determined 7 requirements that would guide our design choices, as shown in **Figure 3.1**. Then in order to understand in more detail how we should design the parts, we decided to create weighing for each criterion. This was achieved by using the pairwise (numerical) comparison matrix. As seen in the **Appendix 3.1** from the matrix, the three most important requirements with all a scores of 21% were aerodynamic performance, structural integrity, and weight, so we had to optimize these aspects in our designs.

We then proceeded by generating design ideas, some of which at the time seemed reasonable and feasible, but as the project moved forward, we understood that these would not be successful. Using the MCDA matrix, we were able to numerically assess how each design meets the weighted criteria, allowing us to compare them in a more objective way. In **Figure 3.2**, we have provided 3 examples of the preliminary designs that we chose not to use in the UAV, because they not only received a lower score in the MCDA matrix but had some concerns.

The rectangular low surface area fairing (design 2) was not optimized when it came to aerodynamic performance. The flow could separate easily due to the sharp edges, and the shape could also potentially interfere with the payload rings, decreasing the ease of assembly. Similarly, a half aerofoil fairing (design 5) did not extend over the entire aerofoil which was not desirable as this would create drag and turbulence. Moreover, this design had very thin edges that could fail during the manufacturing or testing stages. Finally, the aerofoil outlines shaped fairing which also extends the entire chordwise length (design 4), got one of the higher scores in the downselection, but required several improvements as shown in the results of Gate 3, part 8. Initially the design featured a rectangular extrude that would fit between the main wing joint and the top clamp, effectively being attached to the UAV through a boltless clamping mechanism. However, this had to adjusted as removing the fairing in the wind tunnel with this connection mechanism would be difficult. This led us to develop the final fairing design which, was very similar but deviated in shape to this design. It had an additional extrude to extend over the exposed part of the wing spar, had different placement for the FRS attachment, and was connected to the fuselage using bolts.

To conclude, down selection was a valuable tool that allowed us to compare our designs objectively. Having more guidance on the function and role of a fairing would have been helpful in creating more realistic initial designs, which were created from scratch. Additionally, even the final selected design had flaws, like being heavy, as we had to come to comprise between the 7 requirements. Moreover, down selection did not reflect the problems and setbacks related to design that we would face during the project, especially requirements imposed to us by other teams.

Feature or requirement for comparison	Details about that requirement
Cost of materials/processes	Cost of materials used for the chosen design
Structural Integrity	Ability to withstand stresses from false rear spar and aerodynamic forces
Weight	Overall mass of the fairing structure
Ease of assembling	Ease of removal and reassembly
Aerodynamic Performance	Ability to minimize drag and improve the airflow
Manufacturing Complexity	Level of difficulty in production, tool requirements, and lab hours needed for manufacturing
Maintenance Requirements	Frequency and ease of maintaining the fairings

Figure 3.1: Essential criteria for the fairings

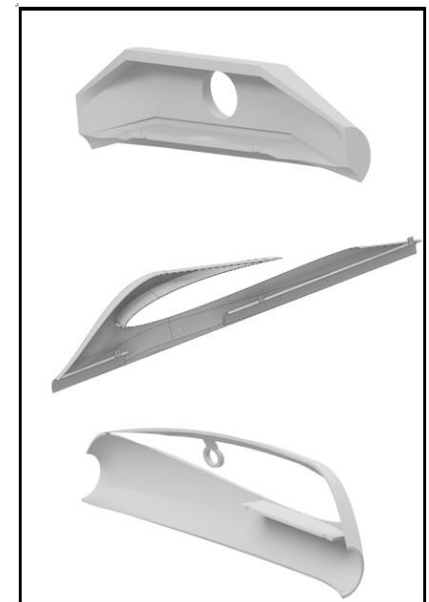


Figure 3.2: Preliminary designs from down selection

3.2 Subassembly Design:

The main objective of our joints were to attach wings, HTP and VTP to the fuselage spar.

The Fus-MWP joint was used to secure the port and starboard wings to the aluminum center bar. As outlined in requirement 2.5.1.a, it needed to keep the wing spars horizontal without inducing any unwanted pitch angles. The diameter of the hole that would hold the center bar in the Fus-MWP joint accounted for tolerances such that the bar would be friction fit. Due to the fact that PLA is not a good load bearing material, an aluminum sheet had to be fit on top of the main wing joint (**Appendix 1.1**), which would hold the shearing loads caused by the bolts. The selection of the center bar and aluminum sheet are detailed in section 4.

The main wing joint clamps connected the joint directly to the CFRP fuselage spar, via M4 bolts at 100 mm spacing. The clamps were locally tapered to account for the change in the fuselage spar to make level. A compromise had to be made when selecting the thickness, as if the clamps were too thin, they could fail and making them thick could lead to a lack of compatibility with the payload. We used two different clamps: one for the structural test, and one for the wind tunnel testing. For the wind tunnel fixtures an additional 8mm counterbore was accommodated to allow for a flanged bronze bush to be inserted (**Appendix 1.2**). The ones for the structural test only had to accommodate for the M4 bolts. A full assembly sequence of the Fus-MWP joint and clamp to the fuselage spar can be seen in **Appendix 1.3**.

The Emp-Fus joint was designed to be aerodynamic so that it would also act as a fairing. Airflow would be turbulent towards the empennage, especially after crossing the main fairing and payload, so creating an additional fairing would only increase the footprint and the drag. The joint was split horizontally into two parts to accommodate for fittings for the VTP and HTP (**Appendix 1.4**). The VTP was attached using glue to the top part and the main and false rear spars of the HTP were held in place with friction fitting (2.5.1.b). The Emp-Fus clamp housed the servo of the empennage team and also secured the joint, via bolts that attached to brass inserts (**Appendix 1.5**) in the top part of the empennage joint. The clamp also accounted for the local taper to allow for the clamp to sit flush with the CFRP fuselage spar.

Moving onto the fairings, the team was responsible for creating smooth and aerodynamic surfaces between, the wings and fuselage, and the empennage to minimize (2.6.1.a). The nose cone fairing was inspired by the rocketry industry, and employed a length to diameter ratio of 3, allowing us to minimize the surface area and streamline the airflow. We also used a fairing to cover the front payload ring which would be exposed. The main wing fairing is the biggest fairing which covers the Fus-MWP joint and 3 of the payload rings, and it reduced the drag because it incorporates a streamline design and minimizes abrupt changes in its contour. Additionally, the fairing uses a pinned joint as shown in Appendix 1, to account for the minimal loads created by the false rear spar (**Appendix 1.6**). An additional FRF attachment was added to increase the structural integrity for the structural testing (**Appendix 1.7**). Furthermore, to meet the requirements from the down selection the fairings were easy to assemble/ disassemble, and had a vertical split for the ½ model testing, alongside with a vertical cutout to accommodate for the wind tunnel mountings. View **Appendix 1.8** for the full assembly sequence of the fairing.

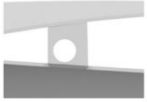
Joints		Fairings	
Part Name	Picture	Part Name	Picture
Fus-MWP Joint		FRF Attachment	
Fus-MWP Joint Clamp		Front payload ring fairing	
Emp-Fus Joint (top)		Hemispherical nose cone fairing	
Emp-Fus Joint (bottom)		Fus-MWP Fairing	
Emp-Fus Joint Clamp			

Figure 3.3: Part name and image of all final designs use on the UAV

3.3 Subassembly Design Reflection

3.3.1 Main wing Joint Reflections

Due to the shape, we selected for center bar joint, the aluminum sheet had small areas where it was not in contact with the surface of the joint (**Appendix 1.1**). This meant that in these areas it wasn't initially load bearing and could cause a failure under larger loads. This could be improved by choosing a different joint shape to make forming the aluminum easier, especially as this joint is covered by the main fairing and therefore its shape doesn't have an impact on aerodynamics.

2.5.2.a 3D printed plastic may be used as a joint fitting former but should not be relied on for significant load transfer. 3D printed formers used for highly loaded joints shall be reinforced with 2014a-T3 alloy sheet straps or



Looking at **Figure 3.4** and requirement 2.5.2.a, you can see where our joint failed, having had to drill holes for the counterbore holes for the wind tunnel attachment left a significant cavity under the wing nut for the structural test. We think this caused the structural failure as the wing nut dropped into the cavity and started to make its way through the thin bit of plastic. We had two types of calmps one for the structural test and one for the wind tunnel mode, but due to lack of time of disassembling the pod we could not have used the clamp designed for the structural load. However, we could use an aluminum plate on the bottom clamp to reinforce and distribute load more evenly over the clamp, giving us more leeway. More on this in the structural review.

Figure 3.4: Damage due to wingnuts on Empennage Clamp

3.3.2 Empennage Joint Reflections

We melted brass inserts into the bottom of the Adaptor (**Appendix 1.4**) due to a lack of space inside of the Adaptor to put the heads of the bolt along with washers in, which was what we had originally planned. While this seemed to work for us here, under larger lateral loads the brass inserts could come loose. As stated, before PLA is not a structural component, and even being more damaged by heating it we were not certain if it could handle relatively big forces. There was no lateral loading in the structural test, but to prevent this from happening we would have to increase the size of the joint to go back to our original idea of having the heads of the bolts inside the adaptor. We could've also used aluminum in some parts as reinforcement, however, this would've required us to change the shape and this joint simultaneously acted as a fairing.

In terms of our Empennage Interfacing, (**Appendix 1.9**) we had integrated the tailplanes seamlessly, we could've improved it even further by adding sleeves for the load-bearing spars to fit into and bearings for the rear spars to rotate in, this would, however, require us to enlarge the Joint. While we had made sure to leave significant room for the rear spars and sanded the contact points to prevent damage to the main spars these would significantly improve the durability of the Empennage in prolonged use and disassembly cycles.

From both these points, it is evident that we should've enlarged our empennage joint slightly, we were only limited by the top plane as it had to match the VTP, other than that we could've made it taller or wider.

3.3.3 Fairings Reflection

While we initially planned on having a separate joint for the false rear spar (**Appendix 1.10**), we had reason to believe that including it in the fairing (**Appendix 1.7**) would be enough as were told by technicians that "unsignificant loads" and therefore would comply with requirement 2.5.2.a. Due to a lack of communication between the design and structures teams we were not able to assemble our fairings properly during the structures test which lead to negative commentary on Gate 4 regarding the compatibility of the fairing. However on March 3 during a trial assembly it was verified that the FRF attachments were aligned with the respective wing spars. In the future it would be valuable to have it assembled correctly to check if a separate joint is a better solution for the FRF integration. Furthermore, we recognize that PLA is not a good load bearing material, so reinforcing the FRF with aluminum would

have been beneficial, even though there were minimal loads. Some difficulties we faced were that both wing teams did not choose the same FRF placements, which lead to an asymmetric fairings and longer designing time.

Weight was another problem that we had with our fairings, while we managed not to impact the weight of the fuselage too much with our joints, our fairings were three times the mass of Company B, this was primarily due to us using 3D print for our fairings, which we had chosen for its ability to print complex shapes, however, had we experimented with different combinations of materials such as a 3D printed exoskeleton covered by film we could've had the best of both worlds.

Finally, the assembly of the fairing was quite complicated itself, having to pass 2 bolts through two holes quite far away from each other while trying to get the fairing to fit well around the fuselage and cover the payload rings takes significant practice to get it done.



Figure 3.4: Verification of compatibility of joints and fairings with full UAV

3.3.4 General Design Reflections

One significant design limitation was our lack of experimentation with different materials or combinations, we committed too early to using only 3D printing due to the many benefits (quick iteration; virtually 0 build time as we didn't have to be present; and its ability to make consistent and accurate complex shapes) and experience we had gained using it for the Main wing joint and Empennage for the graphite GOOSE. Where it cost us was in weight and lack of structural strength, we only realized towards the end, when we started preparing for the inspection in week 16, that our fairings especially were too heavy, however, we were unable to do anything about it as we had to rearrange other joints and fairings which took too much time. This brings us onto a recurring theme through the year - we didn't set strict limits for teams we were working with. This resulted in us having to continuously redesign or adjust parts that we considered that we had "finished", and severely capped our progression rate, we were unable to put many ideas that we had into consideration or prototyping stages, had we restructured our team into 4 design 2 structures, or been less kind, we would've been able to put these ideas into practice.

Halfway through the year because we couldn't continue using our Frankenstein's monster CAD files we redid them from scratch, the main problem we had was the Inventor referencing system was stopping us from continuing to develop our designs. This was because we had been creating assemblies from the different teams' parts and then trying to adjust them, our design process became significantly more streamlined after we had redone the parts using sketches of other teams' CAD files. This was added to the list of our CAD learning outcomes along with, constraining measurements, tolerancing and using a mouse.

3.4 Manufacturing and Integration

Manufacturing is an area where lots of improvement could be done. While the general shapes came out quite well as we had used 3D printing, we had certain problems with the details, the Aluminium plate, and cable ties that we used for one of our fairings was a clear example. We had to print our main fairing as 4 separate pieces, otherwise it wouldn't fit in the 3D printer, and then we glued it together, this was done quite nicely, however, as the 3D printer made an error on one of the edges, we then had to glue a bit of balsa wood to fill in the gap. The bolts and nuts on the fairing were also quite exposed and we should've used brass inserts here also, especially as it isn't a load bearing piece. As one of our payload rings was significantly ahead of the main fairing and made of 3D printed material (the payload was later moved due to weight imbalance), we had to make a relatively large cover for it and attached it with zip ties for simplicity, which was an inelegant solution that made it hard to disassemble. Had we used permanent wires or made the fairing as part of the ring itself we could've achieved a much simpler yet more effective solution.

Moving onto the Empennage, this was done quite well, the brass inserts were melted in quite neatly and maintained a flush fit between the halves, the servo housing was glued onto the clamp quite nicely as well, and the Avionics

Design and Manufacturing Review

teams' cables were also hidden in the channels that we had added to the joint, the biggest issue was the hole for the HTP spar. Having 3D printed the Joint, the printer left material inside the hole that we had to sand out, this wasn't done very well and was even mistaken as a failure during the structural test, had we left an even bigger tolerance this could've been avoided, however, knowing how much material the 3D printer is going to leave in these areas is slightly unpredictable.

As a result of us compromising so much with other teams, integration became our strongest point. Every interface was considered multiple times making sure the fit, shape, tolerance, position suited the other teams existing plans seamlessly, the only exception to this was the previously mentioned False rear spar Joint, which was a result of us not getting in contact with the wing teams fast enough before they cut it. This was further compounded by the member present during the final test being the only one who wasn't present at the practice assembly, meaning a makeshift solution had to be made discarding this joint entirely.

Its important to note that when we first started printing our designs, it was evident that the theoretical dimensions in CAD did not match with the real world prints. To make sure that all the parts were fit smoothly, a tolerance analysis was carried out to get the correct dimensions for CAD. This was done for all of the models that we printed as a team. The example below was for the hemispherical nose cone fairing, and demonstrates how we accounted for the printer uncertainty and the uncertainty of the CFRP fuselage spar.

Table 3.1: Tolerance analysis example for hemispherical nose cone fairing

Questions	Answer (all dimensions in mm)
What tolerance can be achieved on the fairing shaft diameter?	3D printers have ± 0.5 mm uncertainty for the diameter
What is the target clearance?	A minimum clearance of 0.2 mm is needed for compatibility
Minimum width of fuselage inner diameter (from manufacturer of fuselage)?	$36.0 - 0.1 = 35.9$ mm
Maximum permissible shaft size?	Min shaft width – clearance: $35.9 - 0.2 = 35.7$ mm
Nominal shaft size?	Max shaft size – Printer tolerance: $35.7 - 0.5 = 35.2$ mm

3.5 Conclusion

Our biggest failure in this project was the False rear spar joint as it had by far the most issues, the only other piece that failed is the clamps, however, this wasn't so much a design fault, although there are improvements to be made, but a complete oversight in forgetting to print new ones for the structures test. Both these issues can be attributed to time. Due to a lack of time, we were unable to explore different ideas and create more polished solutions, we ended up always being 1 or 2 steps behind the other teams as we kept making adjustments. If we were to redo this year, we would be stricter on our Project management targets and stricter on the teams we were working with and we think this is the solution for fixing our design limitations.

4. Structures

4.1 Review of Structural Features

For the whole main wing plane joint assembly, the load on the top half of the joint was carried by the aluminium sheet which was load bearing. This meant any load on main wing plane joint and top clamp was transferred and carried by the aluminium sheet above protecting the components from excessive loads and damage. However, the bottom clamp does not have to same protections. **(Appendix 2.3)** The forces on the bottom clamp from the bolt can potentially damage the integrity of the part. To rectify this, a load bearing structure can be added to the bottom clamp to transfer the loads. Perhaps another aluminium plate however there are a few considerations to be made. Whether this the best use of our mass allocation as UAV mass is critical and if the damage to this component is minimal then it may best to allow the bottom clamp to be minorly damaged. The forces here should be modelled and calculated to find out if this a necessary structural component. Another point to note is that the space around the bottom clamp is limited because of the pod and payload ring attachments making any sort of reinforcement at this location challenging to fit in the small space available. This leads on to the fact this will add further complexity to an already intricate part that already has an involved assembly not just because of the number of components but also the limited space to work with.

Another potential improvement to our design would be to choose a cuboid centre bar instead of cylindrical one that we selected. There are several benefits on both sides of the argument. For a cylindrical centre there are a greater range of choices for sizing as it needs to fit inside the wing spar so sharing the same cross-sectional shape allows centre bars to be a diameter very close to the diameter of the wing spar whereas this is not the case for a cuboid centre bar and the limit for the length of each side of this component is significantly smaller relatively. **(Appendix 2.4)** The cylindrical centre bar is susceptible to rolling inside the main wing plane joint, which is not ideal for our design. We solved this issue by putting a hole into the centre bar and main wing plane joint for a bolt to go in to hold the centre bar in place to stop rolling. However, this reduced the load bearing capabilities of the aluminium sheet and can reduce the strength and integrity of the part and complicated the manufacturing process. This would not be an issue as the cuboid centre bar doesn't have infinite axis of symmetry allowing it roll about that axis like the cylindrical centre bar does. However, this does mean that calculations for the centre bar bending check becomes more convoluted as when calculating the second moment of area of the cuboid centre bar we now need to consider the orientations of it in our calculation for the reserve factors. This would not be the case with the cylindrical centre bar as it the distance from the centre is the same in all directions for this part, this is only a minor inconvenience and may be worth the hassle.

The empennage joint is made from PLA and has brass inserts bonded inside to slot on top of the fuselage spar. This improves the ease of assembly however the brass inserts can come loose in more extreme cases fortunately in our use cases this is not a problem but is something to note. **(Appendix 2.5)** The empennage itself has no reinforcement on it to handle the loads on it. Fortunately, this is also not an issue as the majority of the loads are transferred to the fuselage spar which is equipped to handle such loads. Also, our load cases are not extreme enough for the transfer of loads to cause any structural deformations at the specified speeds. During the wind tunnel testing the empennage was oscillating a lot. This is not ideal as these vibrations can cause increase the loading and the fatigue stress on the components vibrating and so reduce their operational lifetime. Potential solutions are to increase the stiffness of the components here, so their oscillations aren't so extreme or introduce some damping components perhaps even a TVA system to prevent these extreme vibrations at these load cases.

Clearly mass of the UAV is critical for performance as more mass mean more weight that needs to be lifted in the air. So, optimising and reducing mass is very important for performance. The joints are relatively lightweight relative to what they are made from and because they are key structural components the mass allocation here is necessary. The mass of the fairings, which are not load bearing, for our company were the heaviest out of the 3 companies meaning there is room for improvement. **(Appendix 2.1)** The shape of them is due to aerodynamics so there will only be an evaluation of the material. They were 3D printed using PLA which is not lightweight however this was chosen as design changes kept happening until the last minute, so this versatile method was ideal. Although foam would have been much better in terms of weight, but it has a much longer and complex manufacture process. In hindsight

Structures

there were fairings that did not change much despite the constant redesigns like the nosecone fairing which could have been made from Styrofoam or another lightweight material.

4.2 Comparison of predicted vs experimental results

The predictions for how the joints in the UAV would perform were calculated during TB-1 of the academic year. This was required to be carried out early on so that the design team could perform around the structural requirements of the joints. To begin, the **(Appendix 2.6)** mass table was used to gain a general idea about how much lift was expected to act on the UAV and hence how much force could be expected to act through some of the joints, also the centre of mass was calculated in tandem with this to determine the lift occurring around the empennage joint.

Now that the lift at the main wing joint and the empennage joint had been determined, RF values could be calculated with the safety factors as previously stated. The RF values calculated are as follows:

Figure 4.1: Type of test, part and RF values

Part	Type of test	RF
Spar	Bearing	2.4
Spar	Tension	4.4
Flange	Prying	3.4
Lug	Prying	39.0
Spar	Shear	5.1
Spar	Cleavage	8.9
Pin	Shear	7.4
Centre bar	Bending	2.0

The calculations showed that using a 22.2mm diameter solid bar had an RF of 1.95 (2.0 to 2DP), this was the lowest RF calculated, however these RF checks were rather conservative so the decision to use this centre bar was justified. The prediction was that the centre bar would be the first point to fail in the joints section of the UAV since it had the lowest RF, followed by spar bearing with an RF of 2.4.

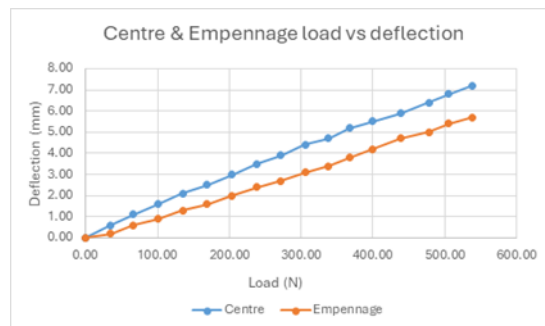


Figure 4.1 Load deflection graph

During the ultimate strength test in TB-2, the UAV was loaded up to 813N, surpassing the ultimate load case. The load vs deflection graph shows the data collected during the strength test and shows a linear relationship between the load and deflection all the way up to the ultimate load case shown in figure 4.1. This confirmed the joints acted in the behaviour predicted from the calculations in TB-1 – demonstrating the validity of Joints and Faring's team as stipulated by the requirement 3.2.5.a. The test carried on after this ultimate load was reached, however the sensors were taken off to prevent damage. After a load of 813N was applied the forces started to decrease to 804N and then suddenly to 764N, this suggested that a yield failure was occurring somewhere in the UAV, however this was not instantaneously obvious, so the UAV was disassembled and inspected afterwards. In figure 4.2 inspection showed deformation in the plastic of the main wing joint clamp, and this along with the data observe led to the conclusion that the bolts in the main wing joint clamp had begun to shear through the plastic of the 3D printed piece.



Figure 4.2: Image of inspection of the main wing joint clamp

This failure was not predicted as no RF values for the 3D printed parts were obtained, and the centre bar was thoroughly inspected for any signs of failure as this is what was predicted to fail – however there was no deformation spotted. Also, to confirm our predictions the rest of the UAV was inspected to see if there were any more deformations hinting to failures. There was a rough lining to some of the holes in the empennage joint, however this was due to the production process and not due to the loads applied to the UAV – indicating that the predictions that were made in TB-1 were correct.

4.3 Critique of analytical and experimental methods

The load cases the calculations are performed for are 6g gusts or manoeuvres so normally our 1g forces would be multiplied by 6. However, in the aerospace industry a safety factor of 1.5 is standard so now the calculations have forces multiplied by 9. This with our assumptions made these calculations rather conservative checks.

To obtain rough values for lift, which was used in preliminary calculations, the centre of mass was calculated to know where to set weight to act. **(Appendix 2.6)** The mass table was used, and these were rough values and were assumptions that were verified by other teams in the company. This was used to calculate the lift on the wing at the main wing plane joint in steady level flight which was approximated into a statics case. This value of lift was used for the majority of the checks.

For the torsion calculations weight was assumed to act 120mm from the leading edge and lift was assumed to act at the back of the wing spar to obtain torsion at the wing root. **(Appendix 2.7)** These assumptions here allowed us to obtain a rough value for torsion.

For the wing loading calculations, the wing spar was approximated to be a cantilever bar with uniform lift and weight across the whole the whole half of the wing spar. **(Appendix 2.8)** This model allowed shear force and bending moment at the wing root to be calculated.

The shear force and bending moment at wing root were modelled to calculate forces along the wing spar. **(Appendix 2.9)** This was combined with the force from torsion on the wing spar to be used for the remaining checks related to the centre bar.

For the lug cleavage check and flange prying check which are checks related to the aluminium sheet we used a different value of lift that we got from our wing teams not the lift from the statics case we modelled which was used in the all the other checks **(Appendix 2.10)**. This discrepancy was only noticed during a final check of the stress report before the data repository freeze. However, this should have been investigated further to see how this happened and which value of lift should have been in this case.

The geometrical properties of the other checks align with the structures slide provided by Ian Farrow (Farrow 2024) **(Appendix 2.11)**.

In terms of experimental methods, there were several errors which may have led to differences in observations between the Graphite Goose test and the Ultimate load test. One issue is that when we tested the UAV in the graphite goose test all the pieces fit together and it acted as we predicted, however during the ultimate strength test we were required to cable tie the structure closed, which then led to deformation being recorded as uneven on each

Structures

wing. This led to the UAV not deforming in the expected manner and ultimately led to a shear failure in the main wing joint clamp.

4.4 Conclusions and Recommendations for Improvements

The structural components designed by the Joints and Fairings team for the UAV were successful. The UAV had great results in the ultimate load test only failing past the design requirement up to 813N. The failure mode, shear along the 3D printed main wing joint clamp, was unexpected showing some shortcomings in the analysis method used here.

The centre bar while successful may have been improved by being using a cuboid shape rather than a cylindrical shape. As the method used while simplified in terms of calculations and had greater variety of options was more complex as a design to implement. The joints for the empennage performed well despite oscillations during tunnel testing. However mass allocation for the UAV could have been optimised better especially the fairings.

A major flaw in analytical methods was using different lift values for the aluminium sheet related checks than for the centre bar relate checks. This inconsistency, which was found late in the project, could have affected the accuracy of the predictions and should have been investigated further. However, as this was a small discrepancy for conservative checks no major issues occurred from this mistake.

For future projects, the main wing plane joint could have been supported by additional load bearing structures under mass and space constraints. Cuboid centre bar designs would reduce complexity without sacrificing structural stability. Empennage oscillations while not extreme enough to cause failures can be reduced through incorporating stiffening or damping parts. Finally mass reduction can be implemented in fairings by using lightweight materials in parts that have low chances of requiring redesigns.

ID	Specification	Date	Initials
3.2.3.b	The pod shall be checked as a beam or truss frame according to your chosen configuration, including checks of frames and longerons, or truss members at ultimate loading for the critical load case.	13/09/24	IF
3.2.3.c	The empennage VTP and HTP main spars and ribs shall be checked under the critical load case, ensuring reasonable deflections at limit.	13/09/24	IF
3.2.4	Wing		
3.2.4.a	The main wing spar and ribs shall be checked under the critical load case, ensuring reasonable deflections at limit. To avoid flutter, compromised flap hinge lines, and compromised mechanism alignment, the wing and flaps shall be checked for excessive deflection or twist.	13/09/24	IF
3.2.5	Joints		
3.2.5.a	The wing root joint, wing to fuselage joint, pod to fuselage joints, and empennage to fuselage joints shall be checked under the critical load case, ensuring all failure modes have satisfactory RF's.	13/09/24	IF
3.2.6	Fuselage and wing testing		
3.2.6.a	To verify loading performance, a representative assembly of fuselage spar, wing spars and HTP spar "Graphite Goose" shall be tested by loading to limit under simulated resultant loads with reaction at forward fuselage cg and empennage spar positions to represent load Case 1). See DRG C5. A fuselage frame and a wing rib and primary joint fittings shall also be nominally tested using bespoke testing schemed by your UAV company. See DRG C6.	13/09/24	IF

Figure 4.3: Image of an extract from the requirements.

5. Aerodynamics

5.1 Port Wing and Flap:

In order to create an XFOIL simulation of the NACA 2414 Port wing, the Reynolds number (Re) and the Mach number (M) were calculated as shown below. It was assumed that the UAV was operating at sea level, such that $\rho = 1.225 [Pa]$, $\mu = 1.789 \cdot 10^{-5} [Pa \cdot s]$, and $a = 343.54 [m/s]$ (Srinivas, 2025).

$$Re = \frac{\rho u c}{\mu} \approx 410844$$

$$M = \frac{V}{a} \approx 0.0582$$

The theoretical behaviour of the aerofoil was plotted for 0° angle of attack with 0° flap deployment and 30° flap deployment as shown in **Appendix 3.1**. Additionally, the experimental plot generated via wind tunnel testing for the corresponding flap deflections have been shown in **Appendix 3.2**.

5.1.1 Theoretical VS Experimental:

Reflecting on the drag polars plots of the theoretical (obtained through XFOIL) and experimental (obtained from the wind tunnel testing) there are evident discrepancies. It can be observed that, the theoretical curves are smoother, exhibit less drag for a given lift coefficient, and the theoretical maximum lift above the experimental data. On the other hand, the wind tunnel data shows greater drag at the same lift level and the maximum peak lift is reduced especially when the flap is deployed.

Comparing the characteristics at 0° flap deflection the theoretical drag polar starts at low C_D and drag gradually rises as C_L increases, reaching its maximum C_L value of 1.25. Now, looking at the experimental curve under the same conditions, the minimum drag is considerably raised, starting with an initial value of 0.03. The peak for C_L has been decreased to 1. Moreover, the curve appears more rounded, meaning that the onset of flow separation occurs earlier accompanied by higher viscous losses.

Raising the flap angle to 30° deflection increases the camber of the aerofoil, thus leading to higher $C_{l_{max}}$ and higher effective angle of attack. In the theoretical drag polar, the peak lift grows such that $C_{l_{max}} = 1.8$, however this comes with the compromise of increased drag. As seen in **Appendix 3.1**, C_D starts at 0.07 extending to 0.15. The experimental drag polar has a $C_{l_{max}} = 1.25$, which is significantly lower than the theoretical, and there is also a significant rise in drag, such that C_D has an initial value of 0.20. All of these were done referring to the repository.

5.1.2 Wind Tunnel Effects:

There are several reasons that could have caused the discrepancies between the theoretical and experimental results beyond the ones discussed above. Firstly, the Reynolds number could deviate from the one assumed in XFOIL, depending on chord length, aerofoil surface conditions and the intensity of the turbulence. Even small Reynolds number differences could shift the laminar to turbulent transition points. Next, it is important to consider the effects of the wind tunnel on the flow. In a wind tunnel, the walls constrict the flow around the aerofoil which consequently increases the velocity, thus changing the pressure distribution. Most importantly, the 2D aerofoil section has an infinite span, however the models employed in the wind tunnel are finite. XFOIL does not consider the minimal tip vortices as well, yet these can potentially reflect onto the experimental results via increased drag and reduced lift.

5.1.3 Conclusion:

The comparison between the theoretical results from XFOIL with the experimental results from the wind tunnel testing for the NACA 2414 Port Wing shows how real world effects can alter the flow and aerodynamic behaviour. While the theoretical drag polar curves showcase smoother behaviour with lower drag, and higher peak lift, the experimental plots depict higher drag with decreased $C_{l_{max}}$ especially when the flap is deflected. Discrepancies in the results are mainly due to the wind tunnel effects and mismatches in the Reynolds number, showing that real world testing is less idealized than the 2D flow model.

5.2 Fairings:

The fairings were newly added to AVDASI 2 this year and as a result there were not any sources to base our designs on. There were no significant equations to calculate the theoretical impact of the fairing. Another alternative way of getting a theoretical representation to evaluate the performance of the fairings was to use CFD analysis, but at our level such a process was not possible due to a lack of technical knowledge, and reliable CFD models, as the commonly available software do not provide credible results. So, there was not a quantitative measure of how well the fairings were expected to work. Hence, as a team we looked at the bigger picture.

Shifting our focus to the company goals, we recognized that the UAV was required to fly as fast as possible to transport blood, such that minimizing the total drag would be the main underlying objective behind our fairings.

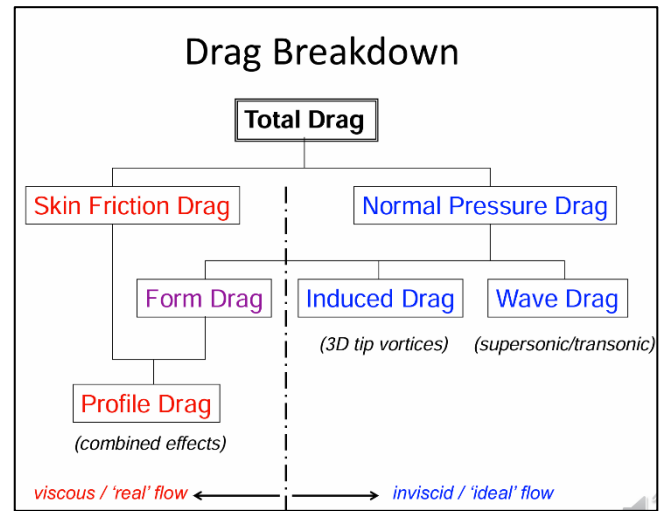


Figure 4.1: Breakdown of drag (Woods, 2023)

Looking at a breakdown of drag, there are two main components: normal pressure drag and skin friction drag. Since the wind tunnel is operating at low speeds of 20 m/s, normal pressure drag was deemed to be negligible. So the main aerodynamic focus was to minimize the form and skin friction drag. It is known that low skin friction drag and low form drag do not tend to go together (Woods, 2023), so our fairing had to have a compromise between increased skin friction drag with lower form drag. Additionally, covering the payload rings wing spars and the main wing joint would be crucial as these were flagged as high sources of drag due to their shape. To give an illustration, based on Sir George Cayley's minimum drag shape (1810) (Woods, 2023), in 2D, a sphere of the same diameter compared to an equivalent airfoil section results in 10 times more drag, meaning that even covering the wing spar (equivalent to a 2D sphere) was important. So, it was necessary to introduce an aerofoil shape into our fairing to cover the exposed part of the wing spar. To complete this goal we used the same shape of the NACA 2414 aerofoil employed by the port and starboard wing teams.

As shown in **Appendix 3.4**, the final design for the fairing that was used in the wind tunnel test was an elongated profile of the wing and a cylindrical body with a streamline shape extending over the main wing joint, clamps and 3 payload rings. Notably, it was designed to maintain laminar flow as long as possible, also the fairing reduces boundary layer thickness, which significantly cuts down on drag. Keeping in mind that these effects would have been multiplied if the UAV had been travelling at the speed under real life operating conditions, however in the wind tunnel test the effects on the UAV were attenuated as the wind tunnel was set to 20 m/s.



Figure 4.2: Exposed aerofoil profile hindering fairing performance

After normalizing the results for the wind tunnel tests as shown in **Appendix 3.3** it became clear that the fairing minimally increased the drag, for the test with fairing on vs fairings off. The main objective of the fairing with cutting drag was compromised due to several factors. First, the fairing was meant to keep the airflow more streamlined on the top of the UAV, however due to the vertical cutout the drag was increased as the airflow interfered between the fairing and the fixtures, visible in **Figure 4.2**. As mentioned before, if the wings were attached to the UAV, without the fairing on, 100 mm of the wing spar would be exposed to the airflow. This would have been a significant contributor to drag due to its cylinder shape. However, due to limitations of the size of the wind tunnel, we were only able to test the fuselage, which implied that we could not quantify this benefit of the fairing, and rather it was reflected negatively, as it was an additional component of the UAV which increases the skin friction. Lastly, the fairing was meant to create a smoother transition for the flow between the UAV body and the wings, however since the wings were not attached the sides of the fairing were open, causing the flow to creep inside and increase the drag.

drastically. Even though all these setbacks were present, after normalizing the drag values, on average the fairings on only recorded 0.09 N greater than the fairings off. This underlines how an increase in skin friction drag lead to a minimal increase in total drag due to highly improved form.

From another perspective, we saw from the wind tunnel data that for fairing 0 angle of attack for some of the trials, the fairing on recorded lower drag than for fairing off. So, this once again shows that the fairing had minimal effect thus hard to decide if it is negative or positive – on the results. These results were expected, as the pod and payload was very large in size, serving as the main source of drag (it can be deduced that), so it overshadowed the effects of the fairing. So, from our point of view the result of the wind tunnel is inconclusive from an experimental point of view, put from a theoretical perspective it is evident that the fairing should reduce drag in the actual operation of the UAV. As mentioned before the wind tunnel does not replicate real world operating conditions. The fairing increases the blockage factor in the wind tunnel, and with the effect of walls, the flow was constricted around the aerofoil which consequently increases the velocity, thus changing the pressure distribution, thus altering the final readings

If the company was more organized a final CAD model assembly could have been provided earlier to our team, which would have allowed us to create a fairing design that is slimmer and has a better integration to the other parts as stated in Gate 4. After speaking with the Unit Director about this, we concluded that in the next years of AVDASI2 using smaller payload rings would have greatly improved fairing performance by lowering its profile and blockage.

Looking at the weight of our fairing, it accounted for 11.85% of the total UAV weight. Since at the moment there is no propulsion, if the UAV were to serve as a glider a heavier overall mass could perhaps be more advantageous. This is analogous to glider competitions, where water ballasts are put in gliders to increase the mass of the gliders so that they perform at higher speeds without increasing the aerodynamic efficiency or the L/D ratio. However, there is more reason to believe that making the fairing lighter would have been more effective in the actual operation of the fairing, as a lighter overall weight means more of the thrust is allocated to carrying the actual payload. So, using lighter materials like aluminium sheets which could have been formed to fit the desired shape could have been better if they were available to us. Or we could have used foam in certain parts of the fairing, with the compromise of jeopardising its real-life operation under humid weather. A quick note on manufacturing though, as the fairing was big and complex in shape, we were obliged to use 3D printing. However this had other disadvantages other than weight as well, like the outer surface of the fairing being rough, especially after certain post processing methods.

The fixture could have been improved, as we failed to accommodate for counterbore for the screws, which lead to the boltheads and nuts being exposed. In actual performance this could be dangerous, and we would have to fix this. To conclude though, a more comprehensive review would have to be carried out about the fairing to come to a fuller understanding of its performance from an experimental perspective.

6. Project Management

6.1 Team and Company Project Plans:

This section evaluates project planning at both a team and company level, focusing particularly on the final design and build phase.

To set up a strong foundation for success, Team 8's Project Managers developed three key planning documents to guide our workflow and ensure readiness for deliverables. Firstly, we made a Work Breakdown Structure (WBS), which decomposed the year-long project into clear, tangible tasks while also assigning responsibilities across the team to specialisms within the team. Next, a Gantt chart was created to provide a visual calendar-like representation of the project timeline, enabling us to track task durations, interdependencies, and deadlines. The benefit of this was that it offered a high-level view of progress and helping us identify bottlenecks before they became critical. For the design and build phase (Weeks 7–17), the approach we took was refined by identifying detailed, feasible tasks and intentionally building in slack time to absorb potential delays and assigning tasks on the Gantt chart to individuals (**Figure 6.1**). This allowed for better progress monitoring and helped mitigate potential scheduling risks. Finally, an Activity-on-Arrow (AoA) network diagram was developed to map out the interdependencies between tasks represented in the Gantt chart. This diagram highlighted the critical path — the longest sequence of dependent tasks that dictates the minimum duration of the project. All three documents are congruent with one another, and this is important as all three documents are representations of the same project from different perspectives, and any misalignments can lead to confusion and miscommunication. They can be found in Appendix Section 4.



Figure 6.1: The Gantt chart from Week 7-17, showcasing dependencies between tasks

6.2 Team Integration into the Company:

Team 8 played an essential role in the UAV design process, as our initial designs predominantly depended on measurements and fittings provided by other teams in the fuselage and wing divisions. This dependence resulted in a challenging design phase beyond the Gate 3 design freeze. For example, there was some delays in the communication of essential dimensions and requirements from the HTP, VTP, and Avionics teams, which required many revisions of the empennage joint, yet the main wing joint and clamps underwent little alterations and were updated with ease. Upon reflection, we saw that proactive engagement with other departments was far more crucial than passive information acquisition. Given our extensive involvement across the company and the fundamental impact of our designs in integrating all the various UAV subassemblies, we knew that our contribution extended beyond our division. Ultimately, our position should be seen as inter-divisional, reflecting our role more accurately.

6.3 Critical Analysis:

Looking back at high-risk elements of our plan, we saw that our critical path did not reflect the actual time that the project took. The critical path indicated that the tasks could all be completed in 48 hours, however as a team we clocked 105 lab hours, excluding time spent on the project outside of the lab. Some of the dependencies were not reflected in the network diagram, as tasks were dependent on the completion of certain designs by other subassemblies as well, leading to unexpected delays. A lack of commitment as a team lead to setbacks, and made us reflect on the dysfunctionalities in our team (Lencioni, 2002). We saw that we only performed during periods of high stress, and during TB1 we did not work to our full ability. This led to underdeveloped designs that would have benefited from a longer iterative process, structures calculations that ran late, and PM documentation that could have included more detail. This was outlined in our team self-assessment below, and in TB2 as a team we were able to address this weakness.

Figure 6.2: TB1 Team self assessment highlighting concerns in areas of commitment and attention to results

Attention to Results	Accountability	Commitment	Healthy Conflict	Trust
3.13	3.43	3.00	3.63	3.75

On a company level, the network diagrams were only specific to divisions and did not account for task float (Appendix 4.4). Furthermore, we observed that as deadlines approached throughout the year, teams increasingly prioritized their individual objectives, becoming more insular. This behaviour was feasible for teams less dependent on external input, unlike ours. Unfortunately, this shift toward individualism diminished overall team performance and limited our ability to achieve more refined and optimized designs. It would have been beneficial to have more communication between teams, especially if the DTMs had constructed the diagrams with more communication with the PMs of their subdivisions. The company level plans should have also included more detail, and should have been documents for guidance that paved the way for success.

We believe that several improvements can be made in the future to avoid the mistakes we faced this year. So, looking at our team level planning documents, we would clearly like to flag dependencies that internal design tasks had on other subassemblies in the company and discuss the PM documents as a team, rather than treating them as deliverables. For example our GANTT chart we would also like to reflect this and implement more milestones besides the gates, as this would serve as additional outlets for feedback. Further optimization could have been done to our GANTT chart by integrating critical paths, and explicitly adding the slack times, especially with tasks that are highly iterative, like the empennage joint. Moreover, assigning tasks to individuals would have increased responsibility and accountability in the team. The overall amount we spent in the lab was substantially greater than the one predicted by the critical path in the network diagram. This could have been due to the fact that the float analysis was not very successful as while we did account for which tasks could be delayed without causing problems within our team, and flagged tasks that had to be completed on time, we did not account for how external effects would impose changes on our work. So, to conclude, we believe that in a way, improving inter-team dependencies in our models as the project managers would have made them more optimized and realistic.

Technology is advancing at such a fast rate, and there are truly a plethora of resources out there to make our lives easier. One particular resource that is commonly used in professional environments are real time collaborative platforms, so similar to the principle of a Gantt chart, platforms like Monday.com allow for real time tracking and communications, and the progress of each team could have been evaluated by the DTMs and TDs. This would have also allowed for the implementation of a dynamic cross-divisional dependency matrix to better reflect how the progress of team was being held back by other, allowing for transparent communication, which would allow in proactive solutions. Hosting specialism related meetings, and using visual traffic-light reporting system to indicate whether the company was on track (green) or critical (red) would have allowed for simply generated yet comprehensive reports to be sent by PMs and TDs.

7. Project Risk Assessment

Our team completed a full risk assessment for Gate 1 of and this document was maintained as a live document to reflect risks that would arise over the course of the project. The goal of this risk assessment was to methodically identify possible hazards, provide clear and effective mitigation plans, improve overall safety and project efficiency. A risk assessment was especially important for Team 8 because of the crucial role that our team had in assuring structural integrity and secure assembly for the whole UAV. Team 8 could not tolerate huge risks because it would directly impact the safety and effectiveness of wind tunnel testing and structural test. Our assessment had a structured risk matrix included severity and likelihood criteria, allowing for optimal prioritizing and strategic allocation of resources to the most significant risks (Appendix 5.2). Furthermore, the assessment underlined the need of attending lab induction sessions and specialized lectures, as well as communicating clearly with lab technicians when uncertainties arose. This made a substantial contribution to upholding safety standards, particularly during actual lab activities.

7.1 Setbacks:

Despite these proactive actions, several identified risks were not adequately addressed. Initially, equipment damage and scheduling were identified as major risks due to their frequency and potential consequences. For example, specific risks, such as risk number 3 in Appendix 5.1, were immediately addressed by emphasizing secure attachment procedures to prevent the wind tunnel from any damage during tests. However, some of the risk assessment's mitigation methods lacked sufficient precision and tangible clarity, diminishing their practical usefulness. Furthermore, the team failed to anticipate certain risks, such as 3D printing failures, due to unfamiliarity with additive manufacturing, but we learned significant lessons from these unexpected obstacles. Furthermore, during structural testing, poor fairing assembly occurred because of missed lab sessions and insufficient intra-team communication. Although the initial risk assessment suggested extensive familiarity with assembly methods, this was not met due to commitment concerns, as noted in the team's self-evaluation.

7.3 Company Level Actions:

On a company level, limited communication between teams, particularly regarding technical issues, resulted in unexpected challenges such as scheduling delays and last-minute revisions during late-stage UAV assembly. Delays in concluding the full-company CAD assembly led to mismatching dimensions during physical assembly, influencing compatibility concerns, resulting in difficulties fitting the plastic payload rings with fairings and ultimately a misaligned centre of mass, resulting in unforeseen design revisions. Furthermore, continual design changes after the official design freeze (Gate 3), along with poorly coordinated slack times among teams, substantially stretched our timetable, reducing overall performance and quality.

7.4 Changes for the Future

Upon reflection numerous key changes were found. A more collaborative and regular updating approach as a company to developing a larger risk assessment might have considerably improved risk detection and mitigation. Furthermore, on a team level, greater involvement from all team members rather than just project managers in risk identification and mitigation planning could have resulted in a more accurate, thorough, and actionable risk assessment, thereby preventing the final joint failure in the structural test. Finally, clearer mitigation methods, improved inter-team communication, proactive scheduling risk management, and collaborative documentation efforts all provide significant chances for improvement. This could have been done by having more specialized meetings, creating more enthusiastic environments and reflecting on our progress would have boosted our outcomes. Taking a different perspective, given that this was the team's first time performing such a thorough risk assessment, it would have been advantageous to work more closely with lab technicians, perhaps getting their help while conducting the risk assessment, as using their knowledge from previous student experiences would have allowed for a more accurate evaluation. Nonetheless, encountering and learning from mistakes is an important part of professional development, demonstrating that not all risks can be completely avoided in a project of this nature.

8. Gate 3- Design Sign-off

Team number: **8**
Sub-assembly: **Joints & Fairings**
Company: **Albatross Aviation**
Assessment date: **13/01/2025**
Overall grading given for Gate 3 (0-100%): **60%**
Assessor: **MG**
Present/absent: **Lianghao Chen absent**

Gate 3 Comments & observations

Comments recorded by the assessor(s).

Strength areas & positive outcomes from the review:

- Good innovation in the centre-spar locating screw

Areas of concern or opportunities/actions for improvement:

- Mounting/securing of empennage seemed unclear
- Some questions not responded to strongly
- MWP fairing design seems very simplistic and the form profile looks to be non-optimum for aerodynamics (although possibly simple is ok). Design appears quite unambitious compared to what might be possible for this new sub-assembly.

9. Gate 4 – Quality Inspection

Team number: **8**

Sub-assembly: **Joints & Fairings**

Company: **Albatross Aviation**

Assessment date: **18/02/2025**

Assessment time and location: 16:00 M005

Overall grading given for Gate 3 (0-100%): **52%**

Assessor(s): MG

Gate 4 Comments & observations

Comments recorded by the assessor(s). Please include photos/evidence where relevant.

Strength areas & positive outcomes from the review:

- Aero profiles fit around wing profile
- Provision made for fitting into tunnel

Areas of concern or opportunities/actions for improvement:

- Seems to have a lack of integration planning with the rest of the UAV
- Design/execution seems a bit ad-hoc/rudimentary
- Cable ties required to close structures

10. Reference List

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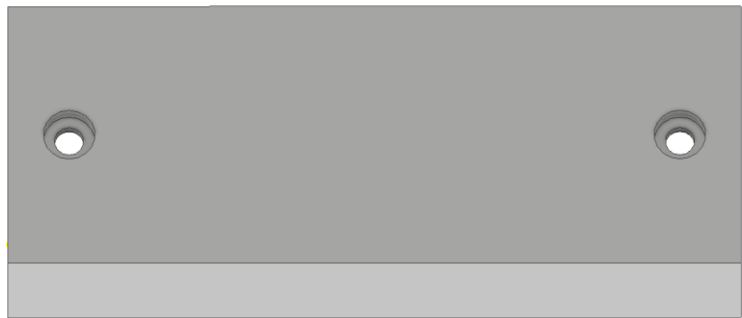
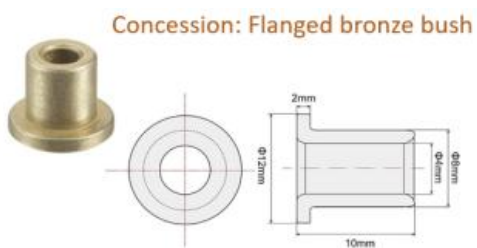
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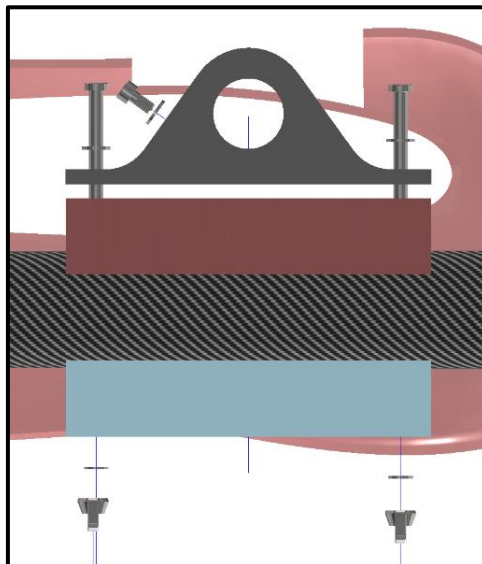
Appendix 1 – Design and manufacturing review



Appendix 1.1: Aluminium sheet used to cover the PLA main wing to fuselage joint to prevent PLA from serving as the primary load bearing surface.

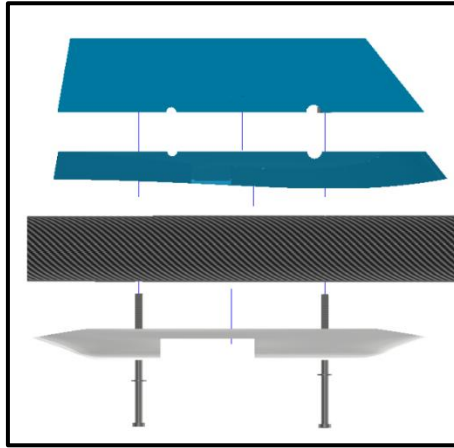


Appendix 1.2: The main wing joint clamp accommodating for the flanges bronze bush by a counterbore of diameter 8mm



Appendix 1.3: The full assembly sequence of the main wing to fuselage joint, alongside with the clamps.

Appendix 1 – Design and manufacturing review



Appendix 1.4: The full assembly sequence of the empennage joint, showing the gaps for the main and false rear spar of the HTP.



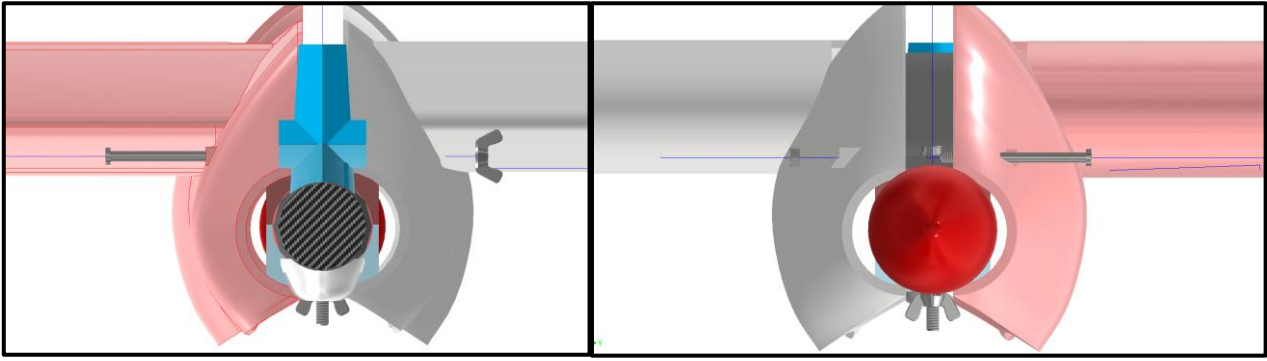
Appendix 1.5: The brass inserts in the top part of the empennage joint to keep the top attached.



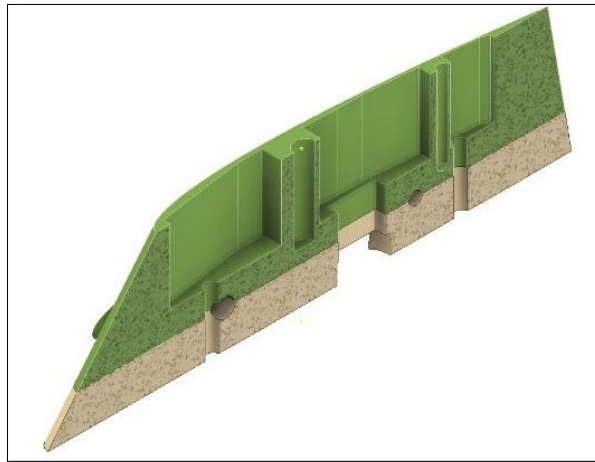
Appendix 1.6: The initial false rear spar pin joint showing clear alignment with the false rear spar of the port wing



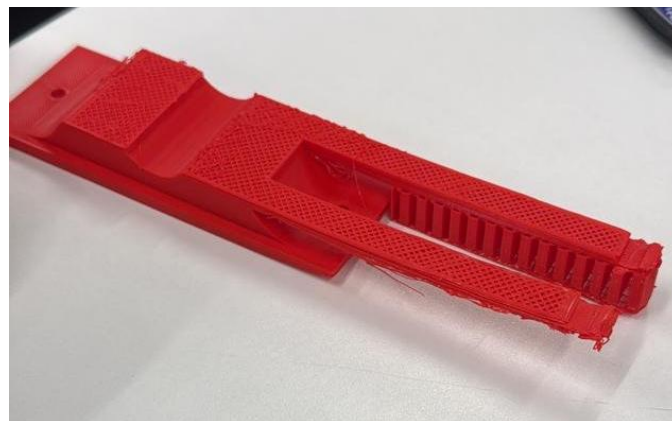
Appendix 1.7: PLA inserts to further reinforce the FRF attachment for the structural test



Appendix 1.8: The assembly sequence of the fairing showing the bolting from the back (left) and front (right)



Appendix 1.9: Cross section view of Empennage Joint



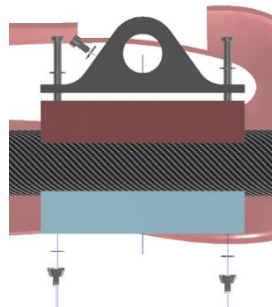
Appendix 1.10: Prototyping of False rear spar joint

Appendix 2 – Structures

Figure 3.7 Weights			Albatross	Bluebird	Cropheed Martin
Assembly	Notes	Units			
Port Wing*	Included avionics.	kg	1.25	1.30	1.40
Starboard Wing*	Included avionics.	kg	1.50	1.30	1.25
Fuselage*	Not including fairings or payload masses.	kg	2.10	2.25	2.70
Fairings†	Fairings only.	kg	0.652	0.212	0.627

Appendix 2.1: Final Weights of Components (Mass Report)

Part	Single Mass (g)	Quantity	Total Mass (g)
Main Wing Plane fuselage joint	68.2	1	68.2
Aluminium Sheet	12.3	1	12.3
M4 x 80mm Bolt	3.1	2	6.2
M4 x 8mm Bolt	1.7	1	1.7
M4 Washers	0.3	4	1.2
Wing Nut	2.0	2	4.0
Main Wing Plane Fairing Half Part	300	2	600
Main Wing Plane Joint Clamp	25.3	2	50.6
Nose Cone Fairing	38.6	1	38.6
Shell Fairing	51.6	2	103.2
Empennage Joint Fairing Upper Half	43.7	1	43.7
Empennage Joint Fairing Lower Half	42.9	1	42.9
M4 x 50mm Bolt	2.8	2	56
Empennage Clamp	32.9	1	32.9
Total			1061.5

Appendix 2.2: Table of UAV Masses for Albatross Aviation**Appendix 2.3: MWP Joint Clamp assembly sequence**

			
15.8	19.0	15.8	22.2
19.0	19.0	19.0	25.4
25.4	25.4	22.2	31.7
	$t = 3.2$		$t = 3.2$

$$\sigma = \frac{My}{I} \text{ where:}$$

$$y = \frac{d}{2}$$

$$I = \frac{\pi(D^4 - d^4)}{64} \text{ for a circle}$$

$$I = \frac{a^4 - b^4}{12} \text{ for a square } a = b$$

$$d = 22.2\text{mm}$$

$$I_{xx} = 11499\text{mm}^4$$

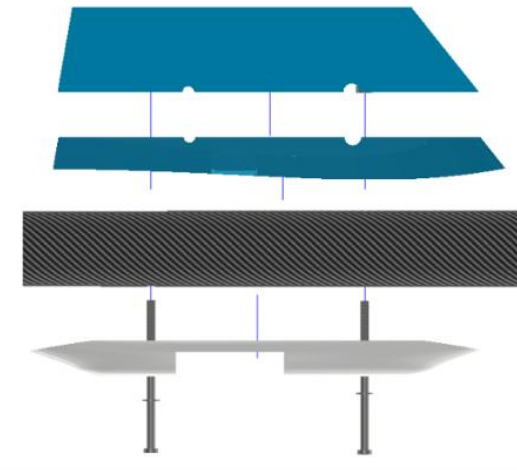
$$y = 11.1\text{mm}$$

$$\sigma = 174\text{Nmm}^{-2}$$

$$\sigma^* = 340\text{Nmm}^{-2}$$

RF = 1.95, for a 22.2mm diameter solid bar.

Appendix 2.4: Centre Bar Selection:



Appendix 2.5: Empennage Assembly sequence

UAV		m	z
		kg	mm
on Fus	Motor	1.000	50
	Batteries	1.000	100
	Payload*	2.500	250
	Fus' tube	0.200	500
	Pod	+sys 0.500	350
	Fus-wing joint	0.050	400
	VTP	+sys 0.150	1250
	Fus-emp joint	0.300	1275
	HTP	+sys 0.300	1300
on Wing	Port wing	+sys 1.000	465
	Batteries	1.000	365
	Star wing	+sys 1.000	465
	Batteries	1.000	365
	Total	10.000	
on Fus		6.000	
on wing		4.000	

2.1. Centre of mass

The data used in this calculation are taken from Table 1.

$$CoM = \frac{m_1 z_1 + m_2 z_2 + \dots}{m_1 + m_2 + \dots}$$

$$\sum m_i z_i = 3.69$$

$$CoM = \frac{3.69}{10} = 0.369$$

Appendix 2.6: Mass Table and steps to get the centre of mass

7.1 Spar shear out check:

$$a = 2d, d = 6mm, t = 2mm$$

$$\tau = \frac{P}{2at} = 34.2 Nmm^{-2}$$

$$\tau^* = 175 Nmm^{-2}$$

$$RF = 5.1$$

7.2 Spar Cleavage Check:

$$a = 2d, d = 6mm, t = 2mm$$

$$\sigma_y = \frac{P}{2at} = 68.3 Nmm^{-2}$$

$$\sigma_y^* = 610 Nmm^{-2}$$

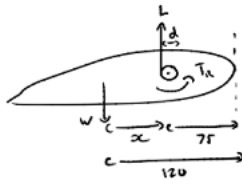
$$RF = 8.9$$

Appendix 2.10: Other RF Checks Examples

Appendix 2 – Structures

3. Torsion

3.1. Wing spar torsion



$d = 30.1 \text{ mm}$
 $W = mg = 2 \text{ g (per wing)}$
 $L = 103.2 \times 0.5 \text{ (per wing)}$
 $= 51.6 \text{ N}$

$$\sum T_R = 0: W \left(120 - 75 + \frac{d}{2} \right) + T - L \frac{d}{2} = 0$$

$$T_R = L \frac{d}{2} - W \left(45 + \frac{d}{2} \right)$$

$$T_R = 103.2 \times \frac{30.1}{2} - ($$

$$T_R = 375 \text{ Nmm at } 1G$$

$$T_R = 3375 \text{ Nmm at } 9G$$

Appendix 2.7: Torsion Calculation



$$a = 50 \text{ mm } b = 50 \text{ mm}$$

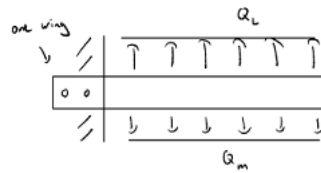
$$\sum M_A = 0: -P_B \times b + M_R + S_R \times a = 0$$

$$P_B = \frac{M_R + S_R \times a}{b} = 3454 \text{ N}$$

$$\sum F = 0: P_B + P_A + S_R = 0$$

$$P_A = -P_B - S_R = -3166 \text{ N}$$

Appendix 2.9: Spar Calculation



$$m_w = 2 \text{ kg}$$

$$Q_L = \frac{P_{fwj}}{s}$$

$$Q_m = \frac{m_w g}{s}$$

$$Q_t = \frac{P_{mwp}}{s} - \frac{m_w g}{s}$$

$$= \frac{P_{mwp} - 4g}{2s}$$

$$S_R = -\frac{P_{mwp} - 4g}{2} = -31.98 \text{ N}$$

at 1g load case.

At 9g, $S_R = -287.82 \text{ N}$

$$M_R = -\frac{P_{mwp} - 4g}{2s} \times \frac{s^2}{2} = 20787 \text{ Nmm}^{-2}$$

at 1g load case.

At 9g, $M_R = 187083 \text{ Nmm}$

Appendix 2.8: Wing Loading Calculation

5.1 Flange Prying Check

Lift from each wing is 195.6N

$$t = 0.5 \text{ mm}, d' = 8 \text{ mm}$$

$$P_{fwj} = 195.6 \times 2 = 391.2 \text{ N}$$

$$\tau = \frac{P_{fwj}}{A_s} \text{ where } A_s = \frac{\pi d'}{4} t$$

$$\tau = 62.3 \text{ Nmm}^{-2}$$

$$\tau^* = 210 \text{ Nmm}^{-2}$$

$$RF = 3.4$$

5.2 Lug Prying Check

Lift from each wing is 195.6N

$$b = 45 \text{ mm}, t = 0.5 \text{ mm}$$

$$P_{fwj} = 195.6 \times 2 = 391.2 \text{ N}$$

$$\sigma_{cl} = \frac{P_{cl}}{A} \text{ where } P_{cl} = \frac{P_{fwj} s}{2} \text{ and } A = bt$$

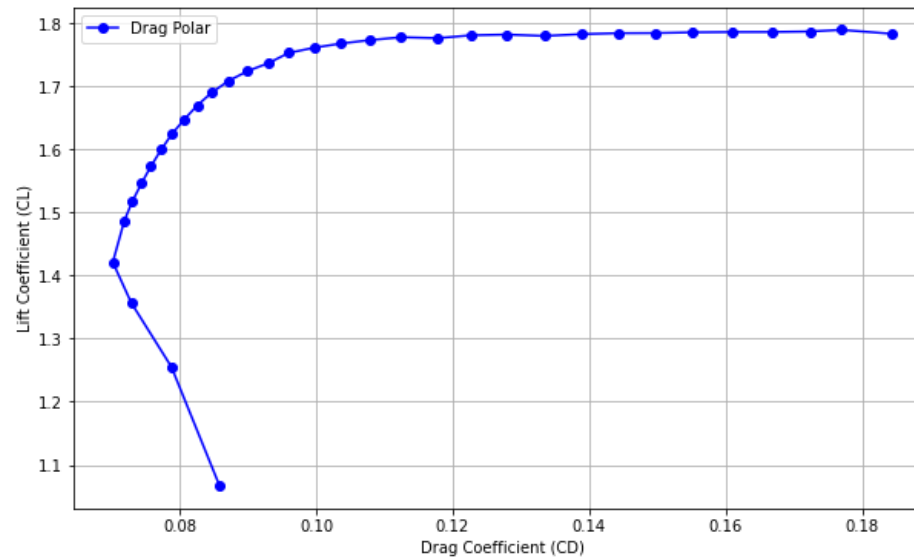
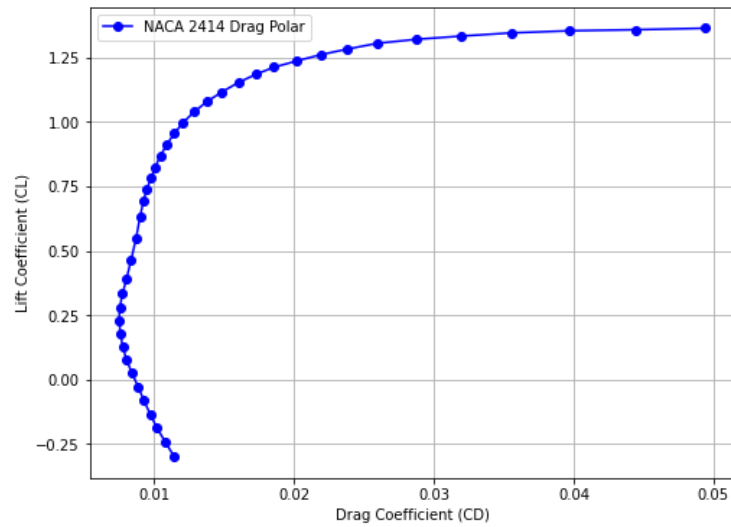
$$\sigma_{cl} = 8.69 \text{ Nmm}^{-2}$$

$$\sigma_{cl}^* = 340 \text{ Nmm}^{-2}$$

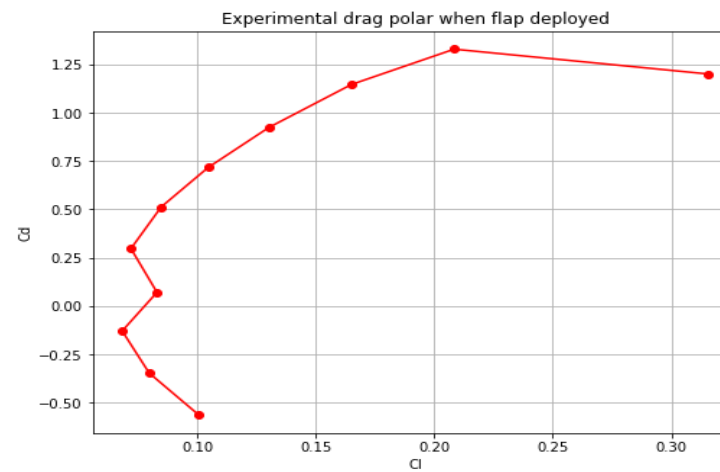
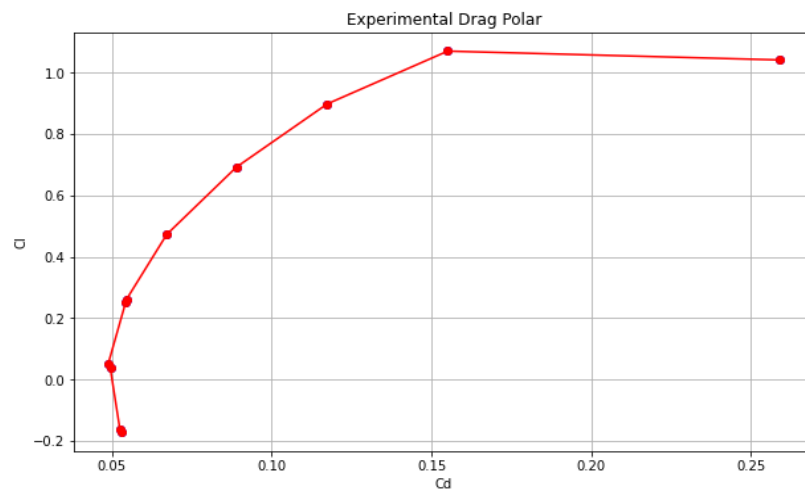
$$RF = 39$$

Appendix 2.10: Aluminium Sheet Checks

Appendix 3 – Aerodynamics



Appendix 3.1: The theoretical drag polar curves obtained for 0° angle of attack with 0° flap deflection and 30° flap deflection

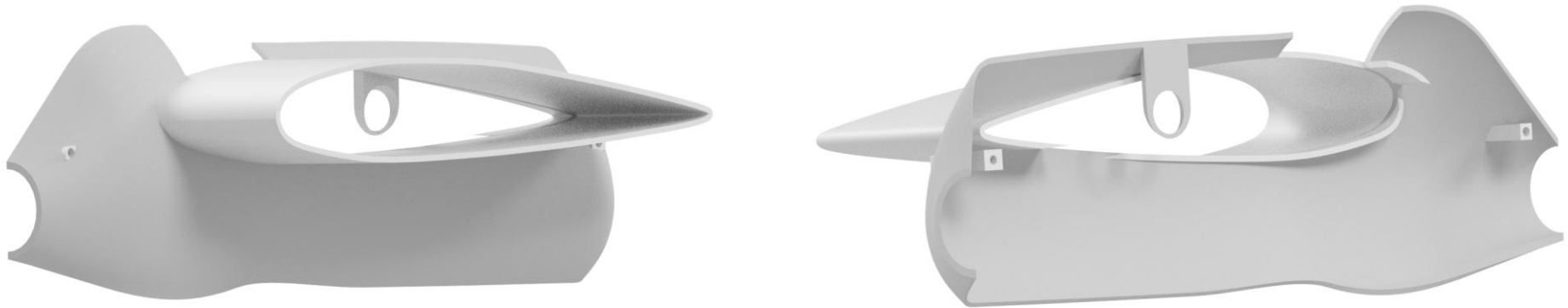


Appendix 3.2: The experimental drag polar curves obtained for 0° angle of attack with 0° flap deflection and 30° flap deflection

Appendix 3 – Aerodynamics

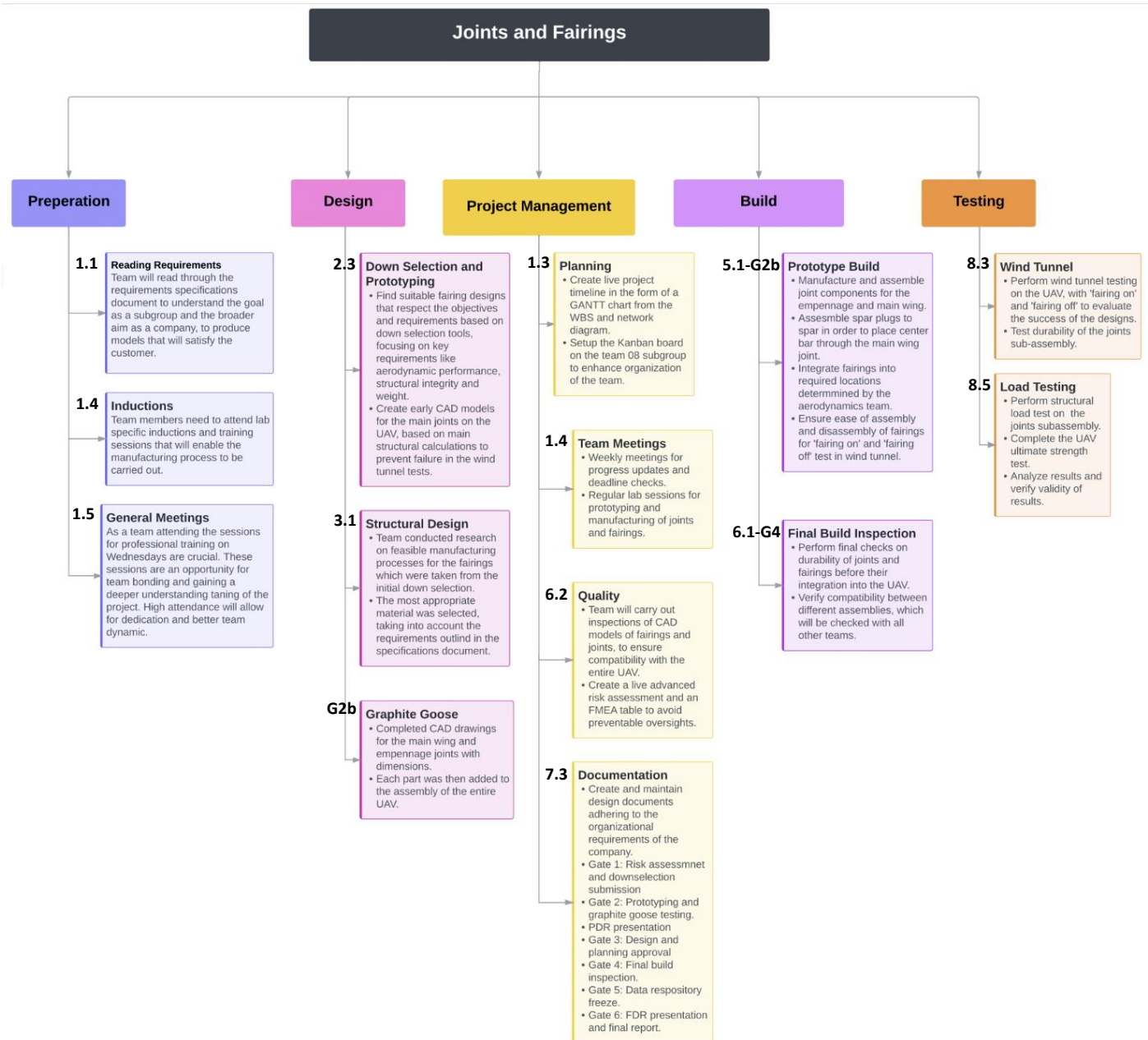
DRAG FX	Drag FX Norm	Lift Fz	Lift Fz N	AoA EXB.Inclination
18.287	4.668	-12.905	-6.467	0
18.407	4.788	-12.747	-6.309	0
18.507	4.888	-12.882	-6.444	0
18.653	5.034	-12.867	-6.429	0
18.949	5.33	-12.781	-6.343	0
19.218	5.599	-12.771	-6.333	0
18.356	4.737	-13.039	-6.601	0
20.855	7.236	-29.177	-22.739	-10
19.128	5.509	-20.398	-13.96	-5
18.669	5.05	-6.357	0.081	5
19.364	5.745	2.169	8.607	10
20.894	7.275	10.021	16.459	15
21.767	8.148	10.563	17.001	16

Appendix 3.3: Table showing drag and lift values obtained in the wind tunnel when the fairings are on



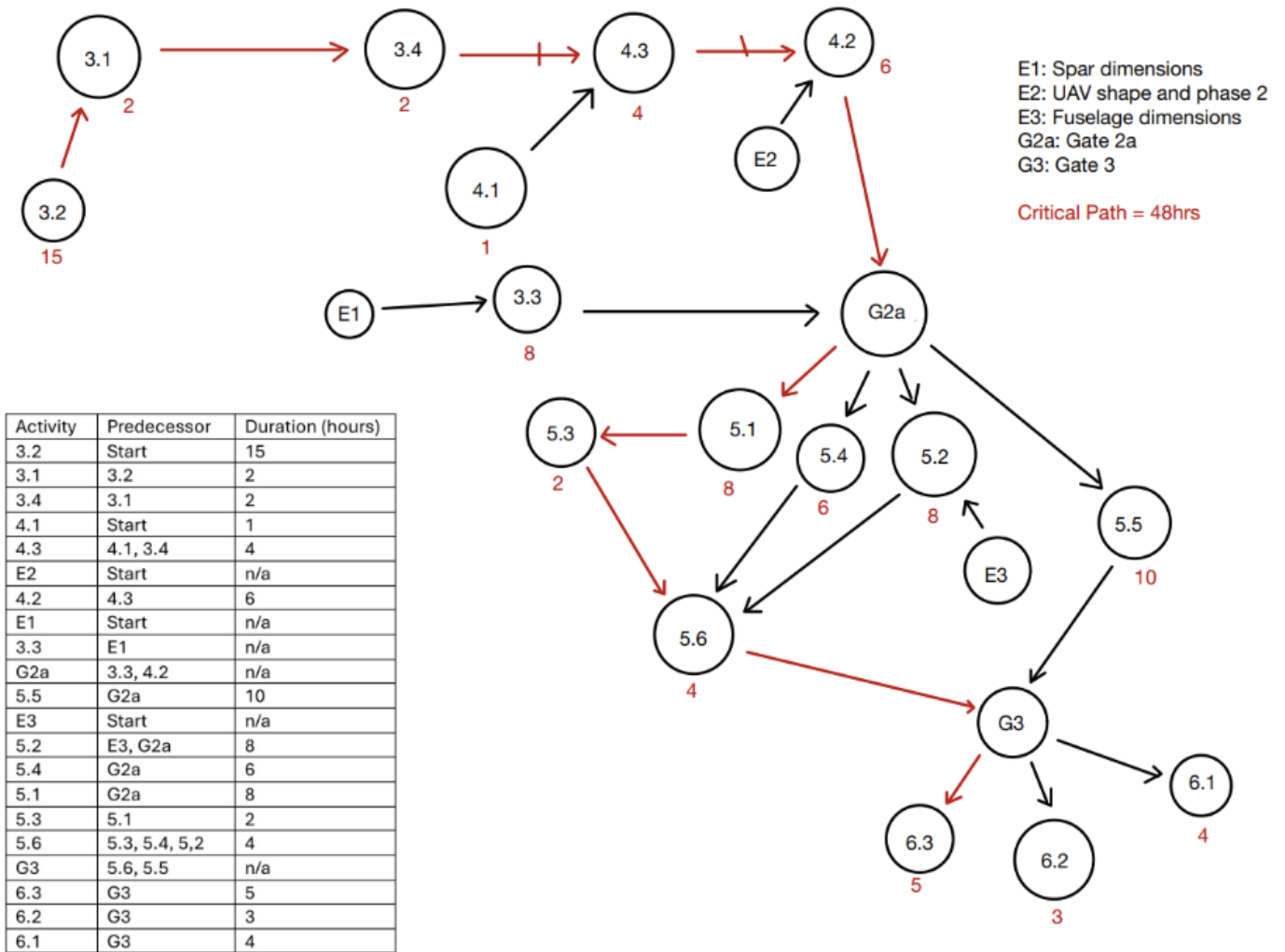
Appendix 3.4: Streamline shape of the main fairing

Appendix 4 – Project Management



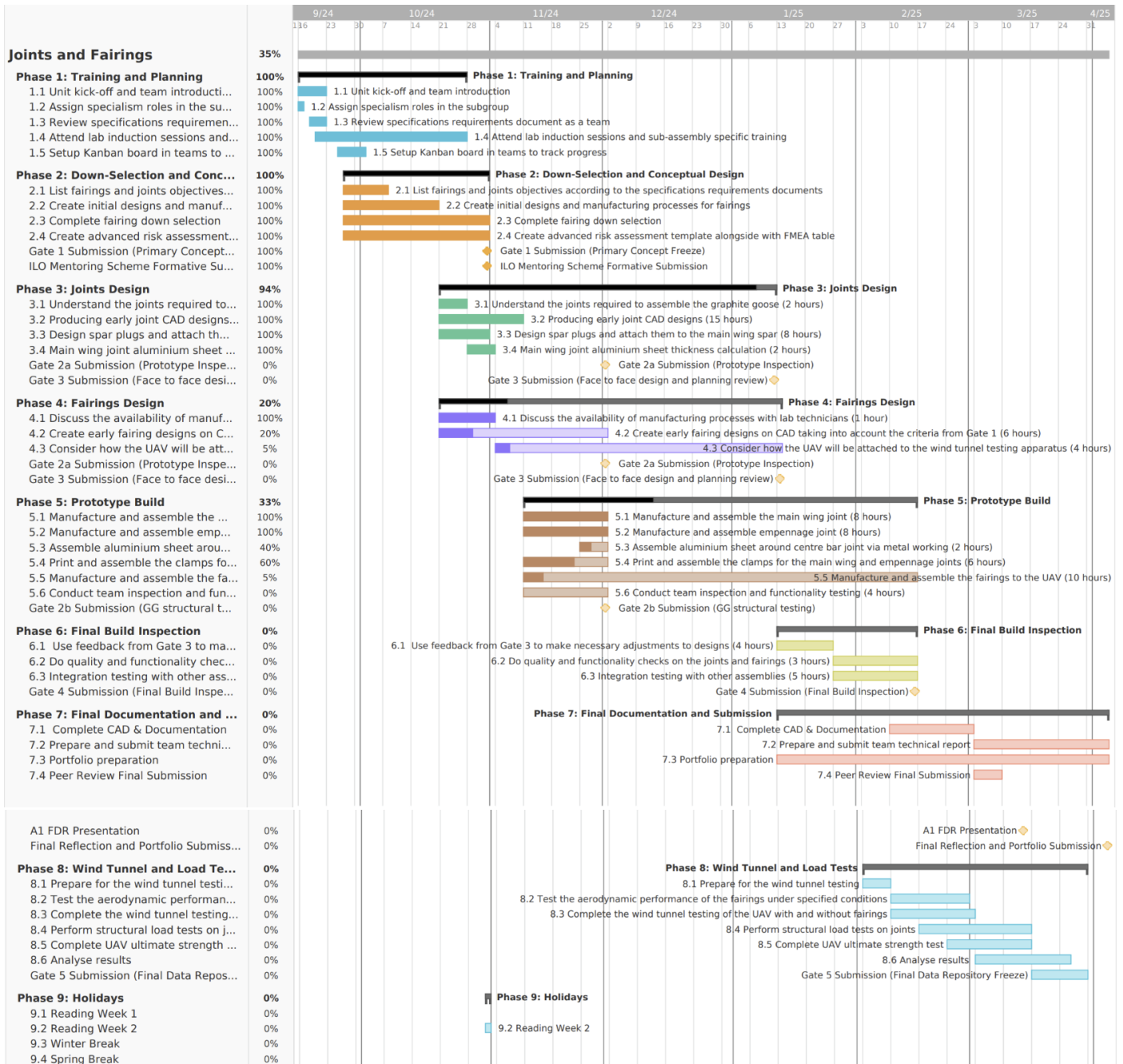
Appendix 4.1: Team 08 Joints and Fairings Work Breakdown Structure. This document visually outlines the planned phases and tasks for the Joints and Fairings subassembly, establishing a structured approach to project delivery.

Appendix 4 – Project Management



Appendix 4.2: Team 08 Joints and Fairings Network Diagram for the design and build phases. This document shows the interdependencies between tasks, and illustrates the critical path based on how long tasks should take.

Appendix 4 – Project Management



Appendix 4.3: Team 08 Joints and Fairings GANTT chart for the entire year. This calendar-like document was used and updated throughout the year by the team

Activity	ID	Predecessor/ Dependencies	Duration (Days)
Place main wing joint	A	-	1
Support the joint with the aluminum sheet	B	A	0.5
Slide in the center bar	C	B	1
Assemble pod and payload to the fuselage body with the rings	D	C	2
Assemble mwj fairing and turtle fairing on top the pod and the joint	E	D	1
Place the nose cone fairing	F	E	0.1
Stabilize the empennage joint on the fuselage	G	F	0.25
Glue the VTP to the emp joint	H	G	0.5
Insert the HTP in the htp	I	H	0.5
Attach Servo Motor and Mechanism to the clamp of the emp joint	J	G	0.5
Put the wires of both VTP and HTP inside the joint	K	G	1
Set all the pressure sensors to their place	L	J, K	2
Skinning for POD and Payload, HTP and VTP	M	I, L	2

Fuselage Assembly Network Diagram

Key:

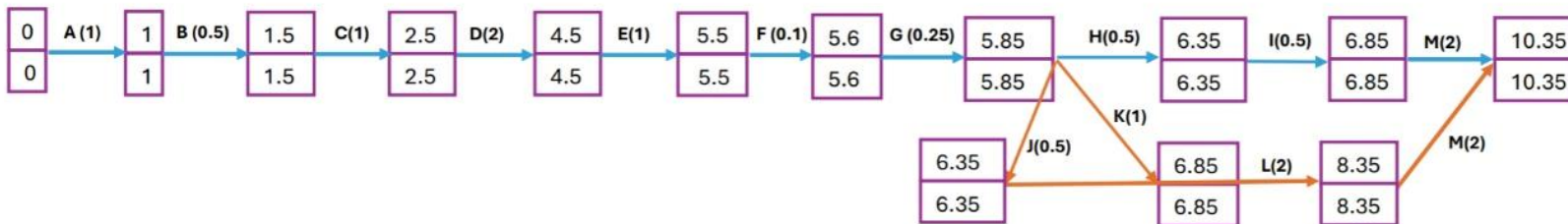
All duration in days

Non-Critical Path 

Critical path 

Earliest start time 

Latest Finish time 



Appendix 4.4: Albatross Aviation Company level network diagrams for the fuselage division. This document provides insight into the inter-team dependencies in this division.

Appendix 5 – Risk Assessment

Risk ID (number)	Type of Risk	Description of Hazard	Description of Consequence or Effect	Pre-mitigation			Mitigations (identify existing and additional)	Post-mitigation			Actions (with owners and date)
				Frequency	Consequence	Risk number		Frequency	Consequence	Risk number	
1-AT-8/10/24	Health and Safety	Improper handling of tools and materials	Potential injuries from sharp objects or power tools	2	4	8	Implement proper safety training, have available safety materials, and have lab technicians on site at all times	1	2	2	Wear gloves, lab coat and safety goggles, alongside with any necessary equipment. Refer to lab safety guidelines and listen to staffs' warnings and instructions. (Week 4)
2-JEB-8/10/24	Scheduling	Delay in manufacturing processes	Project timeline delayed and company gates missed. Pushing back progress to up to a week	3	4	12	Coordination between the team to ensure that a strict timeline is followed, and making sure to book lab slots to avoid manufacturing delays.	1	2	2	Weekly team meetings and progress reports to keep the team in check. Keeping the teams page up to date with deadlines. Every Friday starting 20-10-24.
3-AT-8/10/24	Damage to Equipment/Property	Improper handling of tools and materials	Damage to UAV structure during wind tunnel testing and negative impact on aerodynamic performance. Repairs can range from £10000 for basic repair and up	3	5	15	Periodic quality control checks done by different members of the team to ensure that the assembly meets the requirements, and accurate down selection	2	2	4	Review team designs prior to gates to ensure accuracy and make sure that all members attend their inductions. Carry out every stage, including the preliminary ones with attention and care.
4-JEB-8/10/24	Damage to Equipment/Property	Joint failure during assembly	Material wastage, rework, additional labor costs, and potential delay in project delivery, perhaps costing up to £200	3	4	12	Provide thorough training on joint assembly procedures, use precision tools for assembly, frequent checks	2	3	6	Make sure that members are well informed of the process by attending inductions, and they know the economical consequences for the project. (Week 7)
5-AT-8/10/24	Scheduling	Unavailability of key team members	Delays in project timeline due to key personnel being absent and a need to get them on the same page as the whole team, delaying the completion of the	3	3	9	Have more than one team member with sufficient knowledge in the key specialisms to ensure that delays are avoided and the project is never reliant on one member	1	1	1	Assign more than one student to each specialism and make sure that all members are cross-trained in manufacturing processes in the lab inductions carried out in October 2024.
6-JEB-8/10/24	Damage to Equipment/Property	Over-tightening of fasteners	Damage to joints or fairings, requiring rework or replacement of parts, and potentially damaging other subgroups. Repairs can cost up to £200 and set back	3	4	12	Make sure that team members are well educated in the processes needed to put together the parts that are required through partaking in the lab induction in October 2024.	2	2	4	Team members need to be in constant communication with each other and the DTM to ensure that designs align with each other. Put CAD documents on Teams channel in an organized way so that different teams can access them and design accordingly.
7-AT-8/10/24	Damage to Equipment/Property	Misalignment of joints during assembly	Structural weaknesses in UAV, requiring disassembly and repair, and a redesign of the wing or joint to accommodate for the misalignment	2	5	10	Select materials with high fatigue strength and durability to ensure they are adapted for long-term use. Do not apply loads to joints unnecessarily and store them properly.	1	5	5	Conducting testing to ensure that materials are able to withstand the required loads. This will be verified during the Graphite Goose test.
8-AT-3/10/24	Damage to Equipment/Property	Joints will be fatigued over time	Material fatigue can cause degradation in the structural integrity. Notably, temperature influences the performance of PLA as well.	3	5	15	Discourage team members who aren't mentally present (e.g. sleep deprived, stressed etc.) from doing lab work. Discuss if any physical injuries affect team members to do specific lab work.	1	5	5	Look out for the physical and mental state of team members.
10-VK-13/10/24	Damage to Equipment/Property	Condition of users leading to basic and avoidable errors.	Extensive delays and increase in cost if equipment or parts need maintenance or replacement.	2	5	10	Ensure that team members are thoroughly trained on at least one piece of specialist lab equipment and make sure a lab technician is nearby in case help is required.	1	3	3	Ask other groups that are trained on the specialist equipment on how to use it and don't try to guess.
11-TB-14/10/24	Scheduling	Incorrect use of specialist lab equipment	Could use specialist machinery incorrectly and damage the joints or fairings delaying the production.	4	3	12	Encourage communication between teams, groups and companies so that work towards the UAV can be done efficiently.	1	2	2	Groups on teams and whatsapp are created for PMs and different specialists within groups.
12-TB-14/10/24	Scheduling	Lack of communication within company	within a company and so this could prolong the design process, in turn delaying the manufacturing process and lead to a lower quality UAV.	4	3	12	Make sure all parts have been attached to the fuselage properly and there is no looseness in the attachment. Be prepared for Gate 4. Use bolted connection and zip ties to fix the fairings and joints to the fuselage.	1	5	5	Team members need to be familiar with the assembly sequence and the bolts/parts needed to attach the joint and fairings. Someone from the team will be present at the UAV assembly to make sure that parts are attached correctly.
AT/JEB - 20/02/25	Financial	Loose Parts in assembly	The wind tunnel is a multiple million pound piece of technology and any loose parts like screws or unattached pieces can cause severe damage to the wind	2	5	10	Design iterations alongside with accounting for tolerances in CAD like +0.8 mm for the 3D printers will allow for compatibility with bolts and other subassemblies.	2	3	6	Design specialists will consider tolerances when designing and will check if subassembly is compatible with that of other teams before printing the physical product.
AT/JEB - 20/02/25	Scheduling	Not accounting for tolerances during CAD modelling and other manufacturing processes	This will lead to a setback for the timing of the assembly, as 3D printed parts will need to be adjusted and reprinted to match the requirements which takes around 2 working days to fix, as the CAD	3	3	9	Communicating between teams to have an exact mass of each subassembly and relative positioning on the fuselage spar is important. Measuring each component with a scale to ensure that it meets the weight requirements in the requirement specification document is crucial for company success. It is important to make sure the components are 500g in	1	3	3	As a team we will measure the weight of each of the components that we have manufactured and make them as lightweight as possible.
AT/JEB - 20/02/25	Health and Safety	Center of mass misalignment	backward could lead to instability of the UAV or excessive stability which is not required. Having heavy fairings of or joints that cause this misalignment can lead to safety risks that will be carried into the wind tunnel and structural test.	1	5	5	Encourage teams to only carry the UAV between labs when it is necessary, and to warn PMs to be careful with sensitive components.	1	3	3	Disassemble parts and store in boxes whenever possible to make sure that they are not accidentally damaged.
AT/JEB - 20/02/25	Damage to Equipment/Property	Damage to the UAV during transportation between labs and test sites	Dropping the UAV or hitting it during transportation can lead to serious damage of components. For example, the empennage joint is held together by brass inserts which are melted into the PLA, which cannot withstand forces caused by collisions.	2	4	8	Attend the specialism classes to ensure that there is sufficient knowledge on the procedure to carry out theoretical calculations.	1	3	3	Team needs to do thorough calculations and have quality control over the results by the end of Week 7.
AT/JEB - 20/02/25	Damage to Equipment/Property	Incorrect calculations of structures leading to wrong choices of center bar and aluminum sheet thickness.	The shear stresses from the M4 bolts can rip through the plastic if the wrong aluminum sheet is selected, or the center breaking could not withhold the bending moment exerted by the wings.	2	5	10					

Appendix 5.1: The risk assessment evaluation showing pre-mitigation and post-mitigation risk number alongside with brief descriptions.

Appendix 5 – Risk Assessment

Frequency ratings (how often is this likely to happen?)			Consequence ratings								
			Type of Risk = Health and Safety			Type of Risk = Damage to Equipment/Property			Type of Risk = Scheduling/Financial Risk		
Category Score	Classification Term	Frequency	Category Score	Classification Term	Description/Examples	Category Score	Classification Term	Description/Examples	Category Score	Classification Term	Description/Examples
1	Rare	Extremely unlikely to occur; would only happen under exceptional or extraordinary circumstances.	1	Insignificant	No injury or negligible impact to health, possibly just discomfort or inconvenience.	1	Insignificant	No or negligible damage to equipment or property, easily repaired with no significant cost or impact.	1	Insignificant	No significant impact on the schedule or financials; any delays or costs are negligible and easily absorbed.
2	Unlikely	Not expected to happen frequently, though it may occur under certain unusual conditions.	2	Minor	Minor injury or mild health impact that requires minimal first aid or medical attention, no long-term effects.	2	Minor	Minor damage that requires low-cost repairs, minimal downtime, and no lasting effects on operations.	2	Minor	Minor delays or small financial impacts that cause slight disruptions, but can be managed without affecting overall project outcomes.
3	Neutral	Could happen occasionally, but there is no strong expectation of either frequent occurrence or infrequency.	3	Moderate	Injury that may require medical treatment but is not life-threatening, potential for temporary disability or recovery with treatment.	3	Moderate	Moderate damage that leads to temporary disruption, requires repair or replacement, with moderate financial cost.	3	Moderate	Moderate delays or financial costs that lead to temporary project slowdowns, requiring adjustments but without severe consequences.
4	Possible	Has a fair chance of occurring and may happen under typical conditions, though not guaranteed.	4	Major	Serious injury or health impact that requires hospitalization and could result in long-term disability or extensive recovery time.	4	Major	Major damage resulting in significant downtime, costly repairs or replacements, and potential long-term operational impact.	4	Major	Significant delays or financial overruns that heavily impact the project schedule or budget, potentially requiring a reassessment of timelines and resources.
5	Frequent	Likely to occur regularly and expected in most scenarios or under normal circumstances.	5	Catastrophic	Death or serious injuries to mental and physical health	5	Catastrophic	Complete loss or severe damage to equipment or property, leading to high-cost replacement and long-term disruption or shutdown.	5	Catastrophic	Complete project failure due to extreme delays or massive financial losses, leading to potential cancellation or severe restructuring.

Appendix 5.2: The types of risks and the frequency/consequence rating system used in Team 08 Joints and Fairings.